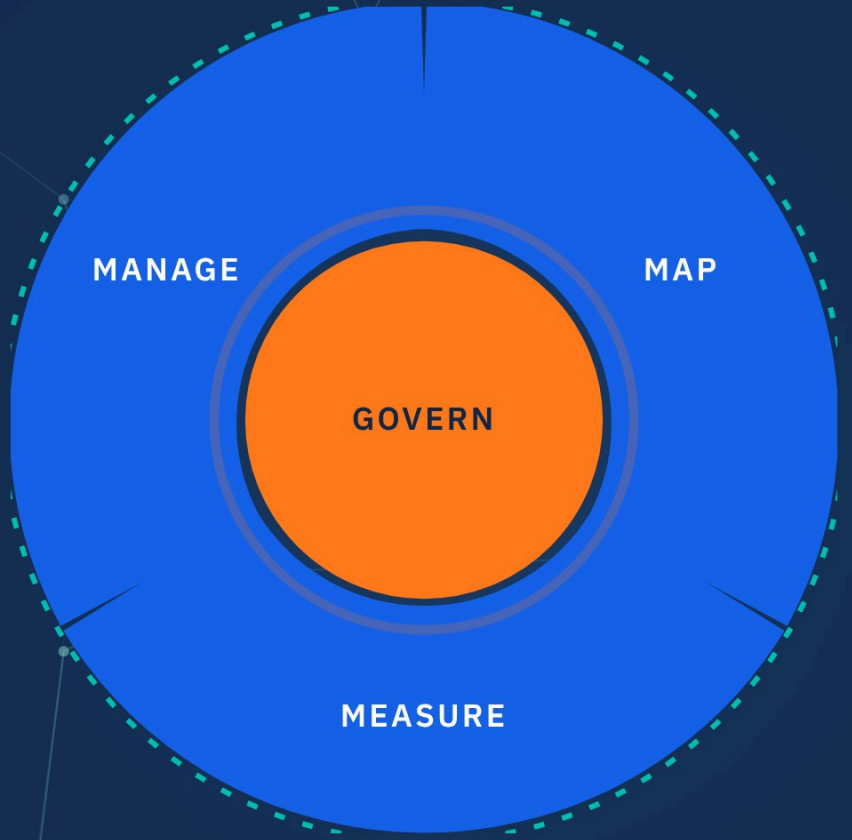
Cyberbiosecurity ÇORUM 2026

DAY 2 · TUESDAY 3 NOVEMBER 2026 · 09:15–10:15

NIST AI RMF

The AI Risk Management Framework: principles and applications for research laboratories



Prof. Dr. Ahmet Altun
Sivas Cumhuriyet University · Faculty of Medicine
Cyberbiosecurity Training Workshop · Hitit University, Çorum

NIST AI 100-1 · AI RMF

AI RISK MANAGEMENT

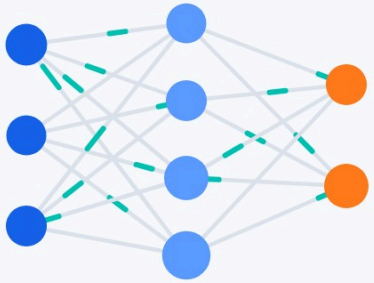
Principles and applications for research laboratories

CYBERBIOSECURITY TRAINING WORKSHOP · ÇORUM

PROF. DR. AHMET ALTUN

FROM CYBER RISK TO AI RISK

Same logic, new kind of system



YESTERDAY · CSF 2.0

Six functions for cyber risk

GV ID PR DE RS RC

TODAY · AI RMF 1.0

Four functions for AI risk

GOVERN MAP MEASURE MANAGE

THE BRIDGE

Both put governance first. The habits you built yesterday (owners, inventory, review cycle) carry straight over.

DUAL USE · DRUG DISCOVERY · 2022

40,000

candidate molecules in under six hours

A model built to steer drug candidates **away** from toxicity was asked to reward toxicity instead. Its output included VX and other known chemical warfare agents.

Nothing was synthesised.

The authors published to warn the field, not to show how.

Urbina F, Lentzos F, Invernizzi C, Ekins S. Nat Mach Intell. 2022;4:189–191. doi:10.1038/s42256-022-00465-9

AI IS ALREADY ON THE BENCH

Six places AI enters a life-science laboratory



Structure prediction

Protein models feed experimental design

`Validity · dual use`

Image analysis

Microscopy, viability and pathology scoring

`Drift · bias`

Language assistants

Protocols, code, reports and reviews

`Confabulation · leakage`

Lab automation

Liquid handlers and cloud labs run by scripts

`Safety · security`

Molecule & sequence design

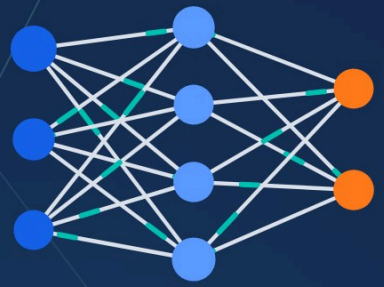
Generative models propose candidates

`Dual use · IP`

Literature mining

Screening papers, extracting data

`Confabulation · coverage`

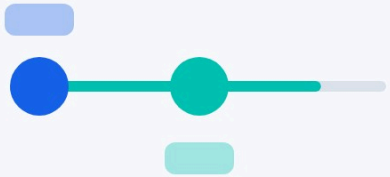


WHY AI RISK IS DIFFERENT · NIST AI 100-1, APPENDIX B

Data changes; models drift.
Behaviour is learned, not written.
Failures are hard to predict.
Errors scale instantly.
The same tool can be used both ways.

THE FRAMEWORK AT A GLANCE

A voluntary framework, still evolving



AI RMF 1.0

NIST AI 100-1, released **26 January 2023**. Voluntary, rights-preserving, not tied to any sector or use case.

PART 1

Framing risk

AI risks and the characteristics of trustworthy AI

PART 2

The Core

Four functions, and how to build profiles

- **JAN 2023**
  AI RMF 1.0 and its companion Playbook
- **JUL 2024**
  Generative AI Profile, NIST AI 600-1
- **MAR 2025**
  Adversarial ML taxonomy, NIST AI 100-2 E2025
- **DEC 2025**
  Cyber AI Profile, NIST IR 8596 (preliminary draft)
- **2026**
  NIST states AI RMF 1.0 is being revised; a critical-infrastructure profile concept note followed in April

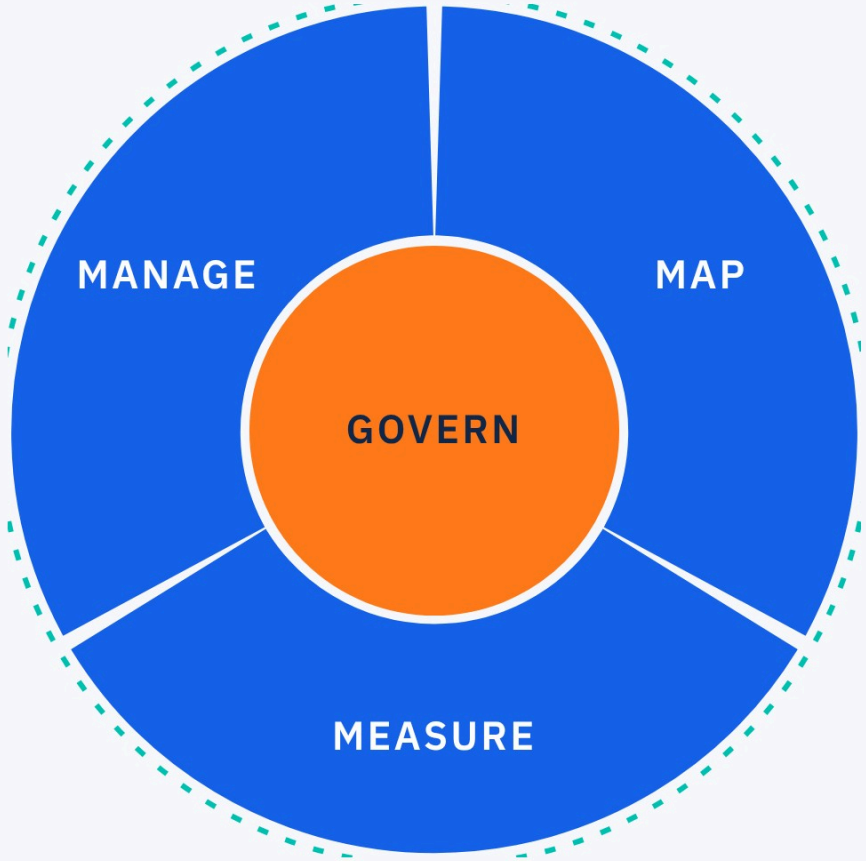
CHARACTERISTICS OF TRUSTWORTHY AI

Seven characteristics, and their trade-offs



THE AI RMF CORE

Four functions; GOVERN runs through all



GOVERN
A culture of risk management is cultivated and present.
6 categories

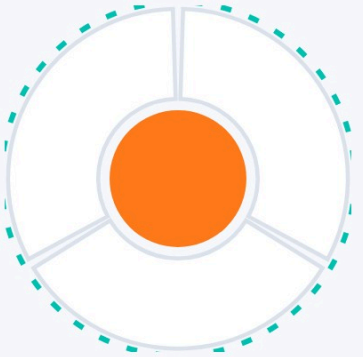
MAP
Context is recognized and risks related to context are identified.
5 categories

MEASURE
Identified risks are assessed, analyzed, or tracked.
4 categories

MANAGE
Risks are prioritized and acted upon based on a projected impact.
4 categories

FUNCTION 1 · CROSS-CUTTING

Govern



A culture of risk management is cultivated and present.

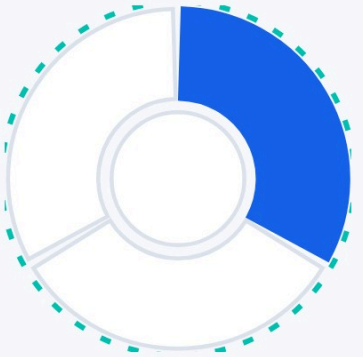
QUESTIONS FOR A RESEARCH LABORATORY

1. Is there a written policy on which AI tools may touch patient, client or genomic data?
2. Who approves a new AI tool in the lab, and who is accountable when it fails?
3. Do we keep an inventory of AI systems in use? (GOVERN 1.6 asks for exactly this.)
4. Are third-party models and services covered by our supplier rules?

START HERE Start with a one-page AI use policy and a register. Most labs have neither.

FUNCTION 2

Map



Context is recognized and risks related to context are identified.

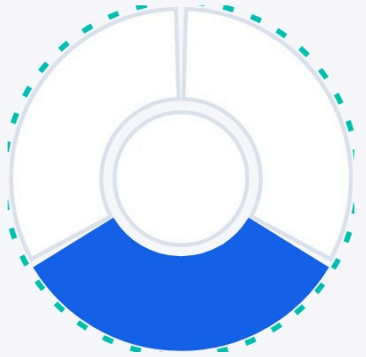
QUESTIONS FOR A RESEARCH LABORATORY

1. What exactly is the system for, and what is it not for?
2. Who uses it, on which data, and who is affected by its outputs?
3. Could the capability be misused, here or by others?
4. Which laws, ethics approvals and client contracts apply?

START HERE Write the intended use down before the first run. Misuse is defined against it.

FUNCTION 3

Measure



Identified risks are assessed, analyzed, or tracked.

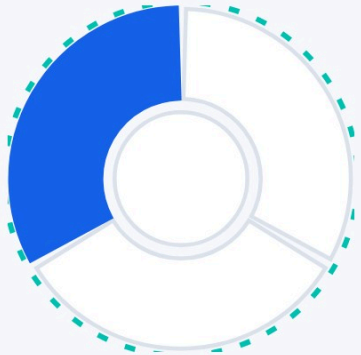
QUESTIONS FOR A RESEARCH LABORATORY

1. Was performance checked on our own samples, not only the vendor’s benchmark?
2. How do we detect drift when reagents, instruments or staining change?
3. Has anyone tried to make it fail on purpose (red-teaming)?
4. Are errors logged and reviewed, with a human able to override?

START HERE Validate locally, then keep measuring. A model validated once is validated for the past.

FUNCTION 4

Manage



Risks are prioritized and acted upon based on a projected impact.

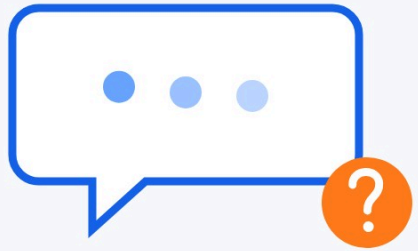
QUESTIONS FOR A RESEARCH LABORATORY

1. Deploy, restrict, or do not deploy: who decides, on what evidence?
2. What is the response plan when the tool produces harmful or wrong output?
3. How are vendor model updates tested before they reach the bench?
4. When is a system retired, and what happens to its data?

START HERE Treat a silent model update like a change of reagent lot: test before use.

GENERATIVE AI PROFILE · NIST AI 600-1 · JULY 2024

Twelve risks made worse by generative AI

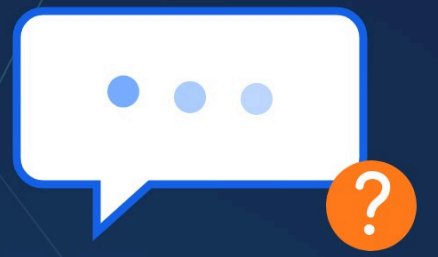


01 **CBRN Information or Capabilities**	02 **Confabulation**	03 Dangerous, Violent, or Hateful Content	04 **Data Privacy**
05 Environmental Impacts	06 Harmful Bias and Homogenization	07 **Human-AI Configuration**	08 **Information Integrity**
09 **Information Security**	10 **Intellectual Property**	11 Obscene, Degrading, and/or Abusive Content	12 **Value Chain and Component Integration**

— Highlighted: the eight that matter most in a research laboratory

CONFABULATION

A fluent answer is not a verified one.



REFERENCES

Plausible citations with invented authors, journals or DOIs

PROTOCOLS

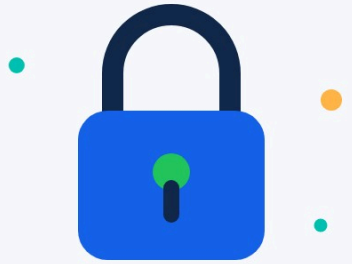
Concentrations, incubation times or biosafety levels stated with confidence and no source

LAB RULE

Every AI-assisted citation is DOI-checked; every protocol step is checked against the SOP or primary source

AI AS THE TARGET · NIST AI 100-2 E2025

The model is also an asset to attack



1

Data poisoning

Corrupt the training or reference data so the model learns the wrong thing

2

Prompt injection

Hide instructions in a document or web page the assistant reads

3

Supply chain

Tampered model weights, packages or plug-ins from third parties

4

Privacy attacks

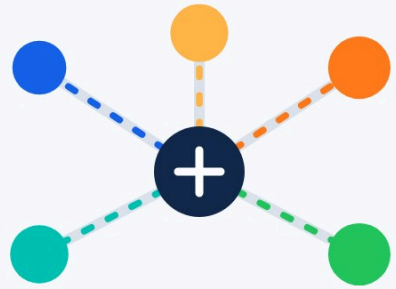
Infer or extract sensitive training data, such as patient records

LINK TO CSF

These map back to yesterday’s CSF functions: PR.DS for data, GV.SC for the supply chain, DE.AE for spotting abuse.

WHERE THE TWO FRAMEWORKS MEET · NIST IR 8596

The Cyber AI Profile applies CSF 2.0 to AI



S

Secure

Securing AI system components

D

Defend

Using AI to strengthen cyber defence

T

Thwart

Countering AI-enabled attacks

STATUS

Preliminary draft, 16 Dec 2025; comments closed 30 Jan 2026. You do not need a separate AI security programme: extend the CSF programme you already have.

WORKED CASE · ILLUSTRATIVE

A cytotoxicity lab adopts an AI classifier



GOVERN	Owner named; tool added to the AI register; use limited to research images, no patient identifiers
MAP	Intended use: viability scoring of MTT and live/dead images; not for regulatory submissions
MEASURE	Compared with two blinded human scorers on 300 local images; re-checked when stain lot or microscope changes
MANAGE	Low-confidence scores go to a human; vendor updates tested on a frozen image set before use

Illustrative design for a pharmacology laboratory; not data from a real study

A MINIMAL AI REGISTER

One table makes GOVERN real



System	Use	Data	Owner	Risk	Human check	Review
Image classifier	Viability scoring	Microscopy images	Lab head	Medium	Low-confidence cases	Each lot change
LLM assistant	Drafting SOP text	No client or patient data	Biosafety officer	Medium	Every step vs. source	Quarterly
Sequence screening	Order checks	Synthetic DNA orders	Core facility manager	High	All flagged hits	Monthly

EXAMPLE ROWS Seven columns, one row per AI system. Start it this week; refine it at the next lab meeting.

BEFORE ANY AI TOOL ENTERS THE LAB

Who owns it?
What data does it see?
What happens when it is wrong?
Could it be turned around?
How will we know it changed?

WHERE THIS GOES NEXT TODAY

The rest of Day 2



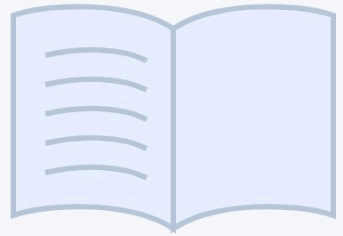
10:30 — **Artificial Intelligence in Life Sciences**
Opportunities, risks and responsible use · Prof. Dr. Gökçe Merey

11:30 — **Institutional Cybersecurity Infrastructure**
A practical case study

14:30 — **Hands-on: apply CSF 2.0 to a biotechnology lab**
Your lab's AI tools belong in that profile too

SOURCES

References and further reading



1. Tabassi E. *Artificial Intelligence Risk Management Framework (AI RMF 1.0)*. NIST AI 100-1; 2023. doi:10.6028/NIST.AI.100-1
2. NIST. *AI RMF: Generative Artificial Intelligence Profile*. NIST AI 600-1; 2024. doi:10.6028/NIST.AI.600-1
3. Vassilev A, et al. *Adversarial Machine Learning: A Taxonomy and Terminology of Attacks and Mitigations*. NIST AI 100-2 E2025; 2025. doi:10.6028/NIST.AI.100-2e2025
4. NIST. *Cybersecurity Framework Profile for Artificial Intelligence*. NIST IR 8596, initial preliminary draft; 16 Dec 2025.
5. Urbina F, Lentzos F, Invernizzi C, Ekins S. Dual use of artificial-intelligence-powered drug discovery. *Nat Mach Intell*. 2022;4(3):189–191. doi:10.1038/s42256-022-00465-9
6. NIST. AI Risk Management Framework (status of revision and profiles); accessed 21 Oct 2026.
7. National Institute of Standards and Technology. *The NIST Cybersecurity Framework (CSF) 2.0*. NIST CSWP 29; 2024. doi:10.6028/NIST.CSWP.29

QUESTIONS & DISCUSSION

Thank you.

Questions on AI risk in the laboratory

Prof. Dr. Ahmet Altun
Sivas Cumhuriyet University · Faculty of Medicine
aaltun@cumhuriyet.edu.tr · ORCID 0000-0003-2056-8683